

Database System Engineering and Distributed Backend Development

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WEEK-2

Problem Statement 1: SkyTrack Airline Management System

Step 1 : Create Database

```
CREATE DATABASE skytrack_db;
```

```
USE skytrack_db;
```

Step 2: Create Flights Table

```
CREATE TABLE Flights (  
    flight_id INT PRIMARY KEY,  
    flight_number VARCHAR(10) UNIQUE NOT NULL,  
    source VARCHAR(50) NOT NULL,  
    destination VARCHAR(50) NOT NULL,  
    departure_date DATE NOT NULL,  
    ticket_price DECIMAL(10,2) CHECK (ticket_price > 0)  
);
```

Step 3 : Create Passengers Table

```
CREATE TABLE Passengers (  
    passenger_id INT PRIMARY KEY,  
    passenger_name VARCHAR(100) NOT NULL,  
    email VARCHAR(100) UNIQUE NOT NULL  
);
```

Step 4 : Create Bookings Table

```
CREATE TABLE Bookings (  
    booking_id INT PRIMARY KEY,  
    passenger_id INT,  
    flight_id INT,  
    booking_date DATE,  
    FOREIGN KEY (passenger_id) REFERENCES Passengers(passenger_id),  
    FOREIGN KEY (flight_id) REFERENCES Flights(flight_id)  
);
```

Step 5 : Insert Records into Flights

INSERT INTO Flights VALUES

```
(101,'AI105','Hyderabad','Delhi','2026-08-10',5500),  
(102,'AI104','Delhi','Mumbai','2026-08-11',4500),  
(103,'AI102','Mumbai','Bangalore','2026-08-12',3500),  
(104,'AI103','Hyderabad','Chennai','2026-08-13',3000),  
(105,'AI101','Chennai','Kolkata','2026-08-14',5000),  
(106,'AI107','Delhi','Hyderabad','2026-08-15',5400),  
(107,'AI108','Mumbai','Delhi','2026-08-16',4700),  
(108,'AI106','Bangalore','Hyderabad','2026-08-17',3300),  
(109,'AI109','Hyderabad','Goa','2026-08-18',4000),  
(110,'AI110','Goa','Delhi','2026-08-19',5200);
```

Step 6 : Display Table

SELECT * FROM Flights;

OUTPUT

flight_id	flight_number	source	destination	departure_date	ticket_price
101	AI105	Hyderabad	Delhi	2026-08-10	5500.00
102	AI104	Delhi	Mumbai	2026-08-11	4500.00
103	AI102	Mumbai	Bangalore	2026-08-12	3500.00
104	AI103	Hyderabad	Chennai	2026-08-13	3000.00
105	AI101	Chennai	Kolkata	2026-08-14	5000.00
106	AI107	Delhi	Hyderabad	2026-08-15	5400.00
107	AI108	Mumbai	Delhi	2026-08-16	4700.00
108	AI106	Bangalore	Hyderabad	2026-08-17	3300.00
109	AI109	Hyderabad	Goa	2026-08-18	4000.00
110	AI110	Goa	Delhi	2026-08-19	5200.00
NULL	NULL	NULL	NULL	NULL	NULL

Step 7: Insert Records into Passengers

INSERT INTO Passengers VALUES

```
(1,'K.Sai Samyuktha','samyuktha@gmail.com'),  
(2,'Chaitra Reddy','chaitra@gmail.com'),  
(3,'Veda Priya','veda@gmail.com'),  
(4,'Charitha','charitha@gmail.com'),  
(5,'Vikram Rao','vikram@gmail.com'),  
(6,'Aditi Gupta','aditi@gmail.com'),  
(7,'Kiran Kumar','kiran@gmail.com'),  
(8,'Hamsika','hamsika@gmail.com'),  
(9,'Rohan Kumar','rohan@gmail.com'),  
(10,'Pooja Nair','pooja@gmail.com');
```

Step 8 : Display Table

SELECT * FROM Passengers;

OUTPUT

passenger_id	passenger_name	email
1	K.Sai Samyuktha	samyuktha@gmail.com
2	Chaitra Reddy	chaitra@gmail.com
3	Veda Priya	veda@gmail.com
4	Charitha	charitha@gmail.com
5	Vikram Rao	vikram@gmail.com
6	Aditi Gupta	aditi@gmail.com
7	Kiran Kumar	kiran@gmail.com
8	Hamsika	hamsika@gmail.com
9	Rohan Kumar	rohan@gmail.com
10	Pooja Nair	pooja@gmail.com
NULL	NULL	NULL

Step 9: Insert Records into Bookings

INSERT INTO Bookings VALUES

(1001,1,101,'2026-08-01'),

(1002,2,102,'2026-08-01'),

(1003,3,103,'2026-08-02'),

(1004,4,104,'2026-08-02'),

(1005,5,105,'2026-08-03'),

(1006,6,106,'2026-08-03'),

(1007,7,107,'2026-08-04'),

(1008,8,108,'2026-08-04'),

(1009,9,109,'2026-08-05'),

(1010,10,110,'2026-08-05');

Step 10 : Display Table

SELECT * FROM Bookings;

OUTPUT

booking_id	passenger_id	flight_id	booking_date
1001	1	101	2026-08-01
1002	2	102	2026-08-01
1003	3	103	2026-08-02
1004	4	104	2026-08-02
1005	5	105	2026-08-03
1006	6	106	2026-08-03
1007	7	107	2026-08-04
1008	8	108	2026-08-04
1009	9	109	2026-08-05
1010	10	110	2026-08-05

Step 11 : INNER JOIN

SELECT

P.passenger_name,

F.flight_number,

F.source,

F.destination

FROM Bookings B

INNER JOIN Passengers P

ON B.passenger_id=P.passenger_id

INNER JOIN Flights F

ON B.flight_id=F.flight_id;

OUTPUT

passenger_name	flight_number	source	destination
K.Sai Samyuktha	AI105	Hyderabad	Delhi
Chaitra Reddy	AI104	Delhi	Mumbai
Veda Priya	AI102	Mumbai	Bangalore
Charitha	AI103	Hyderabad	Chennai
Vikram Rao	AI101	Chennai	Kolkata
Aditi Gupta	AI107	Delhi	Hyderabad
Kiran Kumar	AI108	Mumbai	Delhi
Hamsika	AI106	Bangalore	Hyderabad
Rohan Kumar	AI109	Hyderabad	Goa
Pooja Nair	AI110	Goa	Delhi

Step 12 : COUNT() and GROUP BY

SELECT

destination,

COUNT(*) AS total_flights

FROM Flights

GROUP BY destination;

OUTPUT

destination	total_flights
Delhi	3
Mumbai	1
Bangalore	1
Chennai	1
Kolkata	1
Hyderabad	2
Goa	1

Step 13 : Create Flight_History Table

CREATE TABLE Flight_History (

history_id INT PRIMARY KEY,

flight_id INT,

flight_number VARCHAR(10),

source VARCHAR(50),

destination VARCHAR(50),

action_date DATETIME

);

Step 14: Transaction Example

```
START TRANSACTION;
```

```
INSERT INTO Flights
```

```
VALUES
```

```
(111,'AI111','Hyderabad','Pune','2026-08-20',4200);
```

```
INSERT INTO Flight_History
```

```
VALUES
```

```
(1,111,'AI111','Hyderabad','Pune',NOW());
```

```
COMMIT;
```

Step 15: Create Index

```
CREATE INDEX idx_flight_number
```

```
ON Flights(flight_number);
```

Step 16: Search Using Index

```
SELECT *
```

```
FROM Flights
```

```
WHERE flight_number='AI105';
```

OUTPUT

flight_id	flight_number	source	destination	departure_date	ticket_price
101	AI105	Hyderabad	Delhi	2026-08-10	5500.00
NULL	NULL	NULL	NULL	NULL	NULL

Step 17:Search by Destination

```
SELECT *
```

```
FROM Flights
```

```
WHERE destination='Delhi';
```

flight_id	flight_number	source	destination	departure_date	ticket_price
101	AI105	Hyderabad	Delhi	2026-08-10	5500.00
107	AI108	Mumbai	Delhi	2026-08-16	4700.00
110	AI110	Goa	Delhi	2026-08-19	5200.00
NULL	NULL	NULL	NULL	NULL	NULL

Why is searching by flight_number faster?

Searching by flight_number is faster because each flight has a unique number. The database can quickly find the required flight without checking every record. This saves time and improves the speed of data retrieval, especially when the table contains many records.

Problem Statement 2: MediCare Hospital Management System

Step 1 : Create Database

```
CREATE DATABASE medicare_db;  
  
USE medicare_db;
```

Step 2: Create Doctors Table

```
CREATE TABLE Doctors (  
    doctor_id INT PRIMARY KEY,  
    doctor_name VARCHAR(100) NOT NULL,  
    specialization VARCHAR(50) NOT NULL,  
    consultation_fee DECIMAL(8,2) CHECK (consultation_fee > 0)  
);
```

Step 3: Create Patients Table

```
CREATE TABLE Patients (  
    patient_id INT PRIMARY KEY,  
    patient_name VARCHAR(100) NOT NULL,  
    email VARCHAR(100) UNIQUE NOT NULL  
);
```

Step 4: Create Appointments Table

```
CREATE TABLE Appointments (  
    appointment_id INT PRIMARY KEY,  
    doctor_id INT,  
    patient_id INT,  
    appointment_date DATE NOT NULL,  
    FOREIGN KEY (doctor_id) REFERENCES Doctors(doctor_id),  
    FOREIGN KEY (patient_id) REFERENCES Patients(patient_id)  
);
```

Step 5: Insert Records into Doctors

```
INSERT INTO Doctors VALUES  
(101,'Dr. Amrutha','Cardiology',800),  
(102,'Dr. Veda','Neurology',700),  
(103,'Dr. Harini','Orthopedics',900),  
(104,'Dr. Charitha','Pediatrics',600),  
(105,'Dr. Ravi','Dermatology',800),
```

(106,'Dr. Hamsika','Cardiology',850),
 (107,'Dr. Nandhini','Gynecology',750),
 (108,'Dr. Samyuktha','ENT',650),
 (109,'Dr. Anjali','Ophthalmology',550),
 (110,'Dr. Sameeksha','General Medicine',400);

Step 6: Display Table

SELECT * FROM Doctors;

OUTPUT

doctor_id	doctor_name	specialization	consultation_fee
101	Dr. Amrutha	Cardiology	800.00
102	Dr. Veda	Neurology	700.00
103	Dr. Harini	Orthopedics	900.00
104	Dr. Charitha	Pediatrics	600.00
105	Dr. Ravi	Dermatology	800.00
106	Dr. Hamsika	Cardiology	850.00
107	Dr. Nandhini	Gynecology	750.00
108	Dr. Samyuktha	ENT	650.00
109	Dr. Anjali	Ophthalmology	550.00
110	Dr. Sameeksha	General Medicine	400.00
NULL	NULL	NULL	NULL

Step 7: Insert Records into Patients

INSERT INTO Patients VALUES

(201,'Rajesh Kumar','rajesh@gmail.com'),
 (202,'Neha Sharma','neha@gmail.com'),
 (203,'Priya Reddy','priya@gmail.com'),
 (204,'Pooja Singh','pooja@gmail.com'),
 (205,'Arjun Rao','arjun@gmail.com'),
 (206,'Divya Patel','divya@gmail.com'),
 (207,'Kiran Das','kiran@gmail.com'),
 (208,'Aditi Gupta','aditi@gmail.com'),
 (209,'Nandhini Verma','nandhini@gmail.com'),
 (210,'Anil Kumar','anil@gmail.com');

Step 8: Display Table

SELECT * FROM Patients;

OUTPUT

patient_id	patient_name	email
201	Rajesh Kumar	rajesh@gmail.com
202	Neha Sharma	neha@gmail.com
203	Priya Reddy	priya@gmail.com
204	Pooja Singh	pooja@gmail.com
205	Arjun Rao	arjun@gmail.com
206	Divya Patel	divya@gmail.com
207	Kiran Das	kiran@gmail.com
208	Aditi Gupta	aditi@gmail.com
209	Nandhini Verma	nandhini@gmail.com
210	Anil Kumar	anil@gmail.com

Step 9 : Insert Records into Appointments

INSERT INTO Appointments VALUES

(1,101,201,'2026-08-05'),
 (2,102,202,'2026-08-06'),
 (3,103,203,'2026-08-07'),
 (4,104,204,'2026-08-08'),
 (5,105,205,'2026-08-09'),
 (6,106,206,'2026-08-10'),
 (7,107,207,'2026-08-11'),
 (8,108,208,'2026-08-12'),
 (9,109,209,'2026-08-13'),
 (10,110,210,'2026-08-14');

Step 10: Display Table

SELECT * FROM Appointments;

OUTPUT

appointment_id	doctor_id	patient_id	appointment_date
1	101	201	2026-08-05
2	102	202	2026-08-06
3	103	203	2026-08-07
4	104	204	2026-08-08
5	105	205	2026-08-09
6	106	206	2026-08-10
7	107	207	2026-08-11
8	108	208	2026-08-12
9	109	209	2026-08-13
10	110	210	2026-08-14
NULL	NULL	NULL	NULL

Step 11: INNER JOIN

SELECT

P.patient_name,

D.doctor_name,

D.specialization,

A.appointment_date

FROM Appointments A

INNER JOIN Patients P

ON A.patient_id = P.patient_id

INNER JOIN Doctors D

ON A.doctor_id = D.doctor_id;

OUTPUT

patient_name	doctor_name	specialization	appointment_date
Rajesh Kumar	Dr. Amrutha	Cardiology	2026-08-05
Neha Sharma	Dr. Veda	Neurology	2026-08-06
Priya Reddy	Dr. Harini	Orthopedics	2026-08-07
Pooja Singh	Dr. Charitha	Pediatrics	2026-08-08
Arjun Rao	Dr. Ravi	Dermatology	2026-08-09
Divya Patel	Dr. Hamsika	Cardiology	2026-08-10
Kiran Das	Dr. Nandhini	Gynecology	2026-08-11
Aditi Gupta	Dr. Samyuktha	ENT	2026-08-12
Nandhini Verma	Dr. Anjali	Ophthalmology	2026-08-13
Anil Kumar	Dr. Sameeksha	General Medicine	2026-08-14

Step 12: COUNT() and GROUP BY Report

SELECT

specialization,

COUNT(*) AS total_doctors

FROM Doctors

GROUP BY specialization;

OUTPUT

specialization	total_doctors
Cardiology	2
Neurology	1
Orthopedics	1
Pediatrics	1
Dermatology	1
Gynecology	1
ENT	1
Ophthalmology	1
General Medicine	1

Step 13: Create Doctor_History Table

CREATE TABLE Doctor_History (

history_id INT AUTO_INCREMENT PRIMARY KEY,

doctor_id INT,

doctor_name VARCHAR(100),

specialization VARCHAR(50),

```
registration_date DATE  
);
```

Step 14: Transaction

```
START TRANSACTION;  
  
INSERT INTO Doctors  
  
VALUES  
  
(111,'Dr. Kavya','Oncology',950);  
  
INSERT INTO Doctor_History  
  
(doctor_id, doctor_name, specialization, registration_date)  
  
VALUES  
  
(111,'Dr. Kavya','Oncology',CURDATE());  
  
COMMIT;
```

Step 15: Create Index

```
CREATE INDEX idx_specialization  
  
ON Doctors(specialization);
```

Step 16: Search Using Index

```
SELECT *  
  
FROM Doctors  
  
WHERE specialization='Cardiology';
```

OUTPUT

doctor_id	doctor_name	specialization	consultation_fee
101	Dr. Amrutha	Cardiology	800.00
106	Dr. Hamsika	Cardiology	850.00
NULL	NULL	NULL	NULL